## PLUMBING LEGEND

- DOMESTIC COLD WATER PIPING
- DOMESTIC HOT WATER PIPING
- PIPING(UNDER GROUND)
- DOMESTIC HOT WATER RECIRCULATION PIPING
- VENT PIPING
- WASTE (SANITARY SEWER)
- BALANCING VALVE (BV)
- CHECK VALVE
- PRESSURE REDUCING VALVE
- SHUT OFF VALVE
- DIRECTION OF FLOW
- REDUCER OR INCREASER
- TOP CONNECTION, 45 OR 90 DEGREES
- BOTTOM CONNECTION, 45 OR 90 DEGREES
- SIDE CONNECTION
- CAPPED OUTLET
- DROP IN PIPING
- RISE IN PIPING
- UNION
- SOLENOID VALVE
- THERMOSTAT MIXING VALVE
- PRESSURE GAUGE
- VALVE IN RISER
- WATER METER
- STRAINER

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## PLUMBING ABBREVIATIONS

| | |
|---|---|
| AAV | AIR ADMITTANCE VALVE |
| ABV | ABOVE |
| AFF | ABOVE FINISHED FLOOR |
| A/E | ARCHITECT/ENGINEER |
| ACW | AUTOMATIC CLOTHES WASHER |
| ADA | AMERICANS WITH DISABILITIES ACT |
| ASME | AMERICAN SOCIETY OF MECHANICAL ENGINEERS |
| AST | AUTOMATIC SPRAY TAN UNIT |
| ASTM | AMERICAN SOCIETY FOR TESTING & MATERIALS |
| BHP | BRAKE HORSEPOWER |
| BTU | BRITISH THERMAL UNIT |
| BFP | BACK FLOW PREVENTOR |
| C/I | CONTRACTOR INSTALLED |
| CD | CONDENSATE |
| CONN | CONNECTION |
| CONT | CONTINUOUS |
| CV | CHECK VALVE |
| CW | COLD WATER |
| DFU | DRAINAGE FIXTURE UNIT |
| DIA | DIAMETER |
| DN | DOWN |
| DWG | DRAWING |
| ECO | EXTERIOR CLEAN OUT |
| DW | DISH WASHER |
| °F | DEGREES FAHRENHEIT |
| FCO | FLOOR CLEAN OUT |
| FD | FLOOR DRAIN |
| GAL | GALLONS |
| GPM | GALLONS PER MINUTE |
| HR | HOUR |
| HW | HOT WATER |
| HWR | HOT WATER RETURN |
| HB | HOSE BIB |
| L | LAVATORY |
| MX | MIXING VALVE |
| M.C. | MECHANICAL SUB-CONTRACTOR |
| NFPA | NATIONAL FIRE PROTECTION ASSOCIATION |
| N.T.S | NOT TO SCALE |
| PD | PUMP DISCHARGE |
| PDI | PLUMBING DRAINAGE INSTITUTE |
| PSIG | POUNDS PER SQUARE INCH |
| P.C. | PLUMBING SUB-CONTRACTOR |
| RM | ROOM |
| SAN | SANITARY |
| SH | SHOWER |
| T & P | TEMPERATURE & PRESSURE |
| TPV | TRAP PRIMER VALVE |
| TYP | TYPICAL |
| U/G | UNDERGROUND |
| V | VENT |
| W | WASTE |
| WC | WATER CLOSET |
| WHA | WATER HAMMER ARRESTOR |
| WCO | WALL CLEAN OUT |
| CO | CLEAN OUT |
| WM | WASHING MACHINE |
| TW | TEMPERED WATER |
| WH | WATER HEATER |
| ET | EXPANSION TANK |
| VTR | VENT THROUGH ROOF |

### REFERENCE CODES

1) 2018 GEORGIA BUILDING CODE.
2) 2018 GEORGIA RESIDENTIAL CODE.
3) 2018 GEORGIA MECHANICAL CODE.
4) 2018 GEORGIA PLUMBING CODE.

EQUIPMENT IDENTIFICATION

XXX / X — SYMBOL OR TYPE (SEE SCHEDULE OR SPECIFICATIONS); IDENTIFICATION NUMBER (SEE SCHEDULE)

XX — KEYNOTES

## PLUMBING SPECIFICATIONS

G E N E R A L

RELATED DOCUMENTS:

1. THE GENERAL REQUIREMENTS OF THE ARCHITECTURAL SPECIFICATIONS ARE PART OF THESE SPECIFICATIONS, WHERE AN INCONSISTENCY EXISTS BETWEEN THE WORDING OR INTENT, THIS SECTION SHALL TAKE PRECEDENCE. THE STANDARD FORM OF "GENERAL CONDITIONS" ISSUED BY THE AMERICAN INSTITUTE OF ARCHITECTS DOCUMENT A201, LATEST EDITION, SHALL FORM PART OF THIS CONTRACT.

2. CONTRACT DOCUMENTS MAY INCLUDE, BUT NOT LIMITED TO THE FOLLOWING: ARCHITECTURAL, STRUCTURAL, ELECTRICAL, MECHANICAL, FIRE PROTECTION, AND CIVIL.

DEFINITIONS:

1. TERMINOLOGY: AS DEFINED IN MSS SP-90, "GUIDELINES ON TERMINOLOGY FOR PIPE HANGERS AND SUPPORTS."

2. WATER DISTRIBUTION PIPING: INTERIOR DOMESTIC WATER PIPING.

3. WATER SERVICE: EXTERIOR DOMESTIC WATER PIPING.

4. ACCESSIBLE FIXTURE: PLUMBING FIXTURE THAT CAN BE APPROACHED, ENTERED, AND USED BY PEOPLE WITH DISABILITIES.

PERFORMANCE REQUIREMENTS:

1. DESIGN SUPPORTS FOR MULTIPLE PIPES CAPABLE OF SUPPORTING COMBINED WEIGHT OF SUPPORTED SYSTEMS, SYSTEM CONTENTS, AND COMPONENTS.

2. DESIGN EQUIPMENT SUPPORTS CAPABLE OF SUPPORTING COMBINED OPERATING WEIGHT OF SUPPORTED EQUIPMENT AND CONNECTED SYSTEMS AND COMPONENTS.

3. DESIGN SEISMIC-RESTRAINT (IF APPLICABLE) HANGERS AND SUPPORTS FOR PIPING AND EQUIPMENT, AND OBTAIN APPROVAL FROM AUTHORITIES HAVING JURISDICTION.

4. COMPONENTS AND INSTALLATION SHALL BE CAPABLE OF WITHSTANDING THE WORKING PRESSURE.

SUBMITTALS:

1. CONTRACTOR SHALL SUBMIT FOR REVIEW TO THE ARCHITECT/ENGINEER A COMPLETE LIST OF ITEMS TO BE FURNISHED AND INSTALLED UNDER THIS CONTRACT. INCLUDING, BUT NOT BE LIMITED TO THE FOLLOWING:

   1.1. DIMENSIONED SHOP DRAWINGS OF MATERIALS, FIXTURES AND EQUIPMENT.
   1.2. DIMENSIONED SHOP DRAWINGS OF EQUIPMENT AND PIPING PLAN LAYOUT(S).
   1.3. PRODUCT DATA.
   1.4. FIELD-QUALITY CONTROL INSPECTION AND TEST REPORTS
   1.5. FIELD TEST CERTIFICATES.
   1.6. OPERATION AND MAINTENANCE DATA.
   1.7. EQUIPMENT MANUALS.
   1.8. VALVE TAGS.
   1.9. WELDING CERTIFICATES.

2. ANY DEFERRED CONTRACTOR SUBMITTALS OR CALCULATIONS REQUIRED BY THE LOCAL AHJ, IN ORDER TO OBTAIN OCCUPANCY SHALL BE PROVIDED THE PLUMBING CONTRACTOR. ALL SUBMITTALS SHALL BE PREPARED BY A LICENSED PROFESSIONAL ENGINEER, LICENSED IN THE PROJECT JURISDICTION, AS REQUIRED BY THE LOCAL AHJ. THE CONTRACTORS ENGINEER SHALL PREPARE PLANS, CUT SHEETS, CALCULATIONS, ETC. AS REQUIRED. ALL DEFERRED SUBMITTALS SHALL BE SIGNED AND STAMPED BY THE PREPARING ENGINEER. THE ELECTRICAL CONTRACTOR SHALL INCLUDE ANY FEES REQUIRED FOR SUCH SUBMISSION IN HIS BID.

CODES AND STANDARDS:

1. APPLICABLE CODES:

   1.1. LOCAL PLUMBING CODE.
   1.2. LOCAL DEPARTMENT OF PUBLIC HEALTH.
   1.3. LOCAL AMENDMENTS TO PLUMBING CODE.
   1.4. LOCAL ORDINANCES.

QUALITY ASSURANCE:

1. WELDING: QUALITY PROCEDURES AND PERSONNEL ACCORDING TO ASME BOILER AND PRESSURE VESSEL CODE: SECTION IX.

2. FIRE TEST RESPONSE CHARACTERISTICS: INSULATION AND RELATED MATERIALS SHALL HAVE FIRE TEST RESPONSE CHARACTERISTICS INDICATED, AS DETERMINED BY TESTING IDENTICAL PRODUCTS PER ASTM E 84, BY A TESTING AND INSPECTING AGENCY ACCEPTABLE TO AUTHORITIES HAVING JURISDICTION. FACTORY LABEL INSULATION AND JACKET MATERIALS AND ADHESIVE, MASTIC, AND CEMENT MATERIAL CONTAINERS, WITH APPROPRIATE MARKINGS OF APPLICABLE TESTING AND INSPECTING AGENCY.

   2.1. INSULATION INSTALLED INDOORS: FLAME SPREAD INDEX OF 25 OR LESS, AND SMOKE DEVELOPED INDEX OF 50 OR LESS.
   2.2. INSULATION INSTALLED OUTDOORS: FLAME SPREAD INDEX OF 75 OR LESS, AND SMOKE DEVELOPED INDEX OF 150 OR LESS.

3. PIPING MATERIALS SHALL BEAR LABEL, STAMP, OR OTHER MARKINGS OF SPECIFIED TESTING AGENCY.

4. ASME COMPLIANCE:

   4.1. ASME B31.9 FOR BUILDING SERVICES PIPING VALVES,
   4.2. ASME COMPLIANCE FOR FERROUS VALVES: ASME B16.10 AND ASME B16.34 FOR DIMENSION AND DESIGN CRITERIA.

5. NSF COMPLIANCE:

   5.1. NSF 14, "PLASTIC PIPING SYSTEM COMPONENTS AND RELATED MATERIALS."
   5.2. NSF 61, "DRINKING WATER SYSTEM COMPONENTS - HEALTH EFFECTS; SECTIONS 1 THROUGH 9."

6. ELECTRICAL COMPONENTS, DEVICES AND ACCESSORIES: LISTED AND LABELED AS DEFINED IN NFPA 70, ARTICLE 100, BY A TESTING AGENCY ACCEPTABLE TO AUTHORITIES HAVING JURISDICTION, AND MARKED FOR INTENDED USE.

7. REGULATORY REQUIREMENTS:

   7.1. COMPLY WITH PUBLIC LAW 101-336, "AMERICAN WITH DISABILITIES ACT," AND "UNIFORM FEDERAL ACCESSIBILITY STANDARDS."
   7.2. COMPLY WITH PUBLIC LAW 102-486, "ENERGY POLICY ACT,"

LAWS AND ORDINANCES:

- CONTRACTOR SHALL GIVE ALL NOTICES AND COMPLY WITH ALL LAWS, ORDINANCES, RULES AND REGULATIONS BEARING ON THE CONDUCT OF WORK AS SHOWN ON DRAWINGS AND SPECIFIED HEREIN. IF CONTRACTOR OBSERVES THAT THE DRAWINGS AND SPECIFICATIONS ARE AT VARIANCE THEREWITH, CONTRACTOR SHALL PROMPTLY NOTIFY ARCHITECT/ENGINEER IN WRITING WHEN SUBMITTING BID AND ANY NECESSARY CHANGES SHALL BE ADJUSTED AS PROVIDED IN THE CONTRACT FOR SUCH CHANGES IN WORK, IF CONTRACTOR PERFORMS ANY WORK, CONTRARY TO SUCH LAWS, ORDINANCES, RULES & REGULATIONS, CONTRACTOR SHALL BEAR ALL COSTS FOR CORRECTING THE WORK.

GUARANTEE:

- CONTRACTOR SHALL GUARANTEE WORK TO BE FREE FROM DEFECTIVE WORKMANSHIP AND MATERIALS FOR A PERIOD OF ONE (1) YEAR FROM DATE OF FINAL CERTIFICATE, ANY REPAIRS OR REPLACEMENT DURING THIS PERIOD SHALL BE MADE WITHOUT COST TO THE OWNER, UPON OWNER'S REQUEST.

GENERAL NOTES, FIXTURES, & VALUES

- DESIGN AND CONSTRUCT COMPLETE PLUMBING SYSTEM CONSISTING OF DRAIN. VENT WASTE AND WATER LINES IN COMPLIANCE WITH ALL APPLICABLE STATE LOCAL AND FEDERAL LAWS, CODES AND REGULATIONS. COMPLY WITH ARCHITECTURAL WORKING DRAWINGS FOR LOCATIONS OF ALL PLUMBING FIXTURES.TRIM AND RELATED ITEMS.
- PERMITS AND INSPECTIONS,SHALL BE OBTAINED BY THIS CONTRACTOR AS REQUIRED BY LIMITS OF THE WORK DEFINED HEREIN AND BY APPLICABLE CODES AND REGULATIONS.
- DESIGN DRAWINGS SHALL BE PREPARED BY THIS CONTRACTOR IN COMPLIANCE WITH THE PROJECT DESIGN AND IN COMPLIANCE WITH ALL APPLICABLE CODE REQUIREMENTS. SUBMIT PLUMBING DESIGN ISOMETRIC DRAWINGS TO THE BUILDING DEPARTMENT FOR APPROVAL AS PER THEIR REQUIREMENTS.
- PLUMBING FIXTURES AND FITTINGS SHALL BE INSTALLED IN ACCORDANCE WITH THE CALIFORNIA PLUMBING CODES AND SHALL MEET THE APPLICABLE STANDARDS OF THE CALIFORNIA PLUMBING CODE.
- PLUMBING FIXTURES SHALL BE LISTED IN CONFERENCE WITH CPC 2019.
- HOT MOPPED SHOWER PAN SHALL BE INSPECTED UPON COMPLETION OF HOT-MOPPING AND SHALL BE FILLED WITH WATER FOR INSPECTION. CPC 2019.CIRCULATING HOT WATER SYSTEMS SHALL BE FULLY INSULATED AND SHALL HAVE CONTROL (TIME CLOCK) CAPABLE OF AUTOMATICALLY TURNING OFF THE CIRCULATING PUMP(S) WHEN HOT WATER IS NOT REQUIRED. TITLE 24 SECTION.

WATER CLOSETS, URINALS,SHOWER HEADS, LAVATORY FAUCETS AND SINK FAUCETS SHALL MEET THE FLOWING REQUIREMENTS OUTLINED IN THE TITLE 24 "APPLIANCE EFFICIENCY STANDARDS".

| | |
|---|---|
| WATER CLOSETS | 1.28 GPF |
| URINALS | 0.5 GPF |
| SHOWER HEADS | 1.8 GPM |
| LAVATORIES | 1.2 GPM |
| SINKS | 1.8 GPM |

- FOR ADDITIONS OR IMPROVEMENTS TO A RESIDENCE BUILT BEFORE 1994,EXISTING "NONCOMPLIANT" FIXTURES (TOILETS THAT USE MORE THAN 1.6 GALLONS OF WATER PER FLUSH, URINALS THAT USE MORE THAN ONE GALLON OF WATER PER FLUSH, SHOWERHEADS THAT HAVE A FLOW CAPACITY OF MORE THAN 2.5 GALLONS OF WATER PER MINUTE,AND INTERIOR FAUCETS THAT EMIT MORE THAN 2.2 GALLONS OF WATER PER MINUTE) SHALL BE REPLACED. CERTIFICATION OF COMPLIANCE SHALL BE GIVEN TO THE BUILDING INSPECTOR PRIOR TO FINAL PERMIT APPROVAL. CALIFORNIA SB407.
- SHOWER AND TUB/SHOWER CONTROL VALVES SHALL BE OF THE PRESSURE BALANCE OR THERMOSTATIC MIXING TYPE, HANDLE POSITION SPOTS SHALL BE PROVIDED ON THESE VALVES AND SHALL BE ADJUSTED TO DELIVER A MAXIMUM WATER SETTING OF 120 AS PER CPC 2019.
- PROVIDE ALL VALVES AS REQUIRED. SWEAT COPPER ENDS ON COPPER PIPING SYSTEMS,SCREWED ENDS ON STEEL SYSTEMS. USE SQUARE HEAD COCKS ON FUEL. GAS SYSTEMS,PROVIDE SPEEDWAY STOPS ON ALL FIXTURES.
- WATER PRESSURE REGULATORS SHALL BE INSTALLED ON THE WATER SERVICE PRIOR TO CONNECTION TO BUILDING IF PRESSURE EXCEEDS 80 PSI AS PER CPC 2019.
- HANGERS AND SUPPORTS SHALL BE BY A RECOGNIZED MANUFACTURER SPECIALIZING IN THESE PRODUCTS, SUCH AS MCO, BLINE, ETC
- IN SHOWERS & TUB-SHOWER COMBINATIONS , CONTROL VALVES MUST BE PRESSURE BALANCED OR THERMOSTATIC MIXING VALVES AS PER CPC 2019.
- WATER SYSTEM SHALL BE STERILIZED USING CHLORINE SOLUTION, TESTED AND CERTIFICATION PROVIDED FROM APPROVED TESTING FACILITY.
- ALL SYSTEMS SHALL BE TESTED IN ACCORDANCE WITH THE APPLICABLE SECTIONS OF CALIFORNIA BUILDING CODE AND WITNESSED BY THE LOCAL BUILDING AUTHORITY AND OWNERS REPRESENTATIVE.
- ALL HOSE BIBBS AND FAUCETS FITTED WITH THREADED CONNECTIONS SHALL BE FITTED WITH AN APPROVED NON-REMOVABLE BACK FLOW DEVICE PER CPC 2019
- BUILT-UP SHOWER RECEPTORS SHALL BE INSTALLED AND INSPECTED UPON COMPLETION OF WATER PROOFING AS PER CPC 2019.
- COUNTERSINK PLUG TO BE INSTALLED WHERE RAISED HEADS CAUSE A HAZARD.
- CONDENSATE SHALL BE COLLECTED AND DISCHARGED TO AN APPROVE PLUMBING FIXTURES OR APPROVED DISPOSAL AREA. REFER TO UMC SECTION 310 FOR ADDITIONAL REQUIREMENTS FOR CONDENSATE DRAINAGE PIPING.
- APPROVED SPARK ARRESTORS SHALL BE INSTALLED ON ALL CHIMNEYS AS PER CBC 2019.

WATER HEATERS:

- THE WATER HEATERS TO BE INSTANTANEOUS TYPE.- SEE TITLE 24 CALCS.
- WATER HEATERS SHALL BE EQUIPPED WITH A COMBINATION PRESSURE AND TEMPERATURE RELIEF VALVE AT ALL WATER HEATERS. SET TO OPEN AT NOT MORE THAN 150 PSI. DRAIN PIPE SHALL EXTEND TO OUTSIDE OF BUILDING AND TERMINATE NOT MORE THAN 2 FEET NOT LESS THAN 6 INCHES ABOVE THE GROUND AND POINT DOWNWARD WITH THE END UNOBSTRUCTED. ANY OTHER TERMINATION LOCATION SHALL REQUIRE APPROVAL BY THE BUILDING OFFICIAL AS PER CPC 2019.
- DOMESTIC WATER HEATERS SHALL BE CERTIFIED AND LISTED BY THE CALIFORNIA ENERGY COMMISSION. EXTERIOR INSULATION SHALL BE R-16, OR GREATER. THE FIRST 5 FEET OF EXPOSED HOT AND COLD WATER PIPING AND THE WATER HEATER SHALL BE INSULATED. AS PER TITLE 24.
- PROVIDE A R-12 WATER HEATER BLANKET FOR ALL WATER HEATERS.
- WATER HEATERS LOCATED IN ATTIC OR FURRED SPACE WHERE DAMAGE MAY RESULT FROM LEAKING WATER HEATERS SHALL BE INSTALLED IN OR OVER A WATER TIGHT PAN WITH A 3/4 INCH DRAIN TO AN APPROVED LOCATION AS PER CPC 2019.
- IN SEISMIC DESIGN CATEGORIES D & E WATER HEATER SHALL BE ANCHORED OR STRAPPING SHALL BE AT POINT WITHIN THE UPPER ONE-THIRD (1/3) AND LOWE ONE-THIRD (1/3) OR ITS VERTICAL DIMENSIONS AT THE LOWER POINT. A MINIMUM DISTANCE OF FOUR (4) INCHES SHALL BE MAINTAINED ABOVE THE CONTROLS WITH THE STRIPPING. (CPC 2019),
- ALL HOT WATER PIPES RUN FROM THE HEATING SOURCE TO THE KITCHEN FIXTURES SHALL INCLUDE A MINIMUM OF 1 INCH INSULATION." (CALIFORNIA ENERGY CODE)
- PROVIDE MINIMUM R-12 INSULATION FOR SMALL STORAGE WATER HEATERS WITH AN INPUT RATING LESS THAN OR EQUAL TO 75,000 BTU/HR. (CALIFORNIA ENERGY CODE)
- THE ENTIRE LENGTH OF RE-CIRCULATING DISTRIBUTION SECTIONS OF DOMESTIC WATER SYSTEMS SHALL BE INSULATED, REGARDLESS OF WHETHER THE PIPING IS BURIED OR EXPOSED," (CALIFORNIA ENERGY CODE)
- REQUIREMENT FOR GAS WATER HEATER INSTALLED TO SERVE INDIVIDUAL DWELLING UNITS ES150.0(N)

a. A CONDENSATE DRAIN INSTALLED NO HIGHER THAN 2" ABOVE THE BASE OF THE HEATER THAT ALSO ALLOWS FOR GRAVITY DRAINAGE
b. THE "B" VENT INSTALLED IN A STRAIGHT POSITION (NO BENDS) FROM THE ROOM CONTAINING THE WATER HEATS TO THE ROOF TERMINATION (FOR FUTURE POSSIBLE SLEEVING FOR HIGH EFFICIENCY HEATER VENTING).
c. A 120 VOLT RECEPTACLE ACCESSIBLE TO THE HEATER INSTALLED WITH TO 3'-0".

PIPING:

- INSTALLATION OF ALL PIPING SYSTEMS SHALL BE IN ACCORDANCE WITH APPLICABLE LAPMO INSTALLATION STANDARDS AND LISTED IN CPC APPENDIX 1.
- SOLDER SHALL BE LEAD FREE COMPLYING WITH THE PROVISIONS OF FEDERAL CLEAN DRINKING WATER ACT AND THE CPC 2019.
- 50/50 SOLDER JOINTS AREA NO LONGER PERMITTED IN PORTABLE WATER DISTRIBUTION SYSTEMS. ONLY 95/5 SOLDER JOINTS ARE PERMISSIBLE.
- INFORMATIONAL,FOR THE PURPOSES OF THIS SECTION ,"LEAD FREE" MEANS NOT MORE THAN 0.2 PERCENT LEAD WHEN USED WITH RESPECT TO SOLDER AND FLUX AND NOT MORE THAN 8PERCENT WHEN USED WITH RESPECT TO PIPES AND PIPE FITTINGS WITH RESPECT TO PLUMBING FITTING AND FIXTURES, "LEAD FREE" MEANS NOT MORE THAN 4 PERCENT BY DRY WEIGHT.
- SUPPORT ALL LINES BY MEANS OF RESILENT FITTING OR ACCESSORIES, DO NOT ALLOW PIPING TO BE IN DIRECT CONTACT WITH FRAMING.
- NO COPPER PIPING SHALL BE IN CONTACT WITH ANY METAL OTHER THAN COPPER.
- PIPING SYSTEMS SHALL BE SECURELY SUPPORTED TO PREVENT UNDUE MOVEMENT AND SWAYING BUT FLEXIBLE ENOUGH TO PERMIT EXPANSION.
- ALL DOMESTIC HOT WATER PIPING TO HAVE THE FOLLOWING MINIMUM INSULATION INSTALLED: PIPE(1 INSULATION), 1"PIPE (1" INSULATION); 1/2" PIPE (1/2" INSULATION) 3/4"PIPE (1" INSULATION)1" TO 1-1/2" PIPE (1-1/2" INSULATION),(CPC 2019)
- ADDITIONALLY,THE 1/2" HOT WATER PIPE TO THE KITCHEN SINKED THE COLD WATER PIPE WITHIN 5' OF THE WATER BOTH REQUIRE 1' MINIMUM ISULATION ES 150.0(J).
- BELOW GRADE HOT WATER PIPING IS REQUIRED TO BE INSTALLED IN A WATERPROOF AND NON -CRUSHABLE SLEEVE OR CASING THAT ALLOWS FOR REPLACEMENT OF BOTH THE PIPING AND INSULATION PLEASE NOTE OR SHOW THIS ON THE PLANS.

- DRAIN PIPING SHALL SLOPE AT 2% OR 1/4 INCHES PER FOOT FOR BELOW 4" PIPING UNLESS OTHERWISE NOTED. 4' AND LARGER PIPING REQUIRING SLOPE 1% OR 1/8 INCHES PER FOOT. THE INSTALLATION CONDITIONS SHALL BE NOTED ON THE PLANS AND APPROVAL RECEIVED FROM THE ADMINISTRATIVE AUTHORITY PER CPC 2019.
- COMPENSATE SHALL BE COLLECTED AMP DISCHARGED TO AN APPROVED DISPOSAL AREA. REFER TO CPC 2019 FOR ADDITIONAL REQUIREMENTS FOR COMPENSATE DRAINAGE PIPING.
- F.M,C,O, SHALL OPEN IN A DIRECTION OPPOSITE TO THE FLOW OF THE WASTE ABOVE PIPE F.L
- PROVIDE MINIMUM CLEARANCE OF NOT LESS THAN 18" IN FRONT OF CLEAN OUT.
- ALL SOIL WASTE AND VENT LINES SHALL COMPLY WITH UNIFORM PLUMBING CODE, SOIL AND WASTE BEYOND THE FIVE FOOT BUILDING LINE SHALL BE VITRIFIED CLAY PIPE WITH SPEED SEAL JOINTS OR TRANSIT SEWER PIPE WITH RING GROOVE COUPLINGS.
- COLD AND HOT WATER LINES ABOVE GRADE SHALL BE TYPE M HARD DRAWN COPPER WITH 50-50 SOLDER JOINTS. COLD WATER LINES BELOW GRADE SHALL BE TYPE L COPPER WITH 95-5 SOLDER JOINTS FROM WATER METER.
- FLOOR DRAINS AND FLOOR SINKS WHICH ARE SUBJECT TO INFREQUENT USE SHALL BE PROVIDED WITH AN APPROVED AUTOMATIC METHOD OF MAINTAINING TRAP SEALS AS PER CPC 2019.
- A MINIMUM OF 12" ACCESS PANEL TO BATHTUB TRAP CONNECTIONS SHALL BE INSTALLED WHEN SLIP JOINT CONNECTIONS ARE USED AS PER CPC 2019.
- NO GAS PIPING SHALL BE INSTALLED UNDERGROUND BENEATH ANY BUILDING SLAB ON GRADE OR STRUCTURE UNLESS INSTALLED IN A GAS TIGHT CONDUIT. INCLUDING PIPING UNDER COVERED PATIOS [CPC 2019]
- NO GAS PIPE SHALL BE INSTALLED UNDER THE BUILDING O PATIO COVER
- UNDERGROUND PLASTIC GAS PIPING SHALL BE 18" DEEP AND THE TRANSITION RISER MUST COMPLY WITH CPC 2019.
- AN ELECTRICALLY CONTINUOUS CORROSION-RESISTANT TRACER WIRE (MINIMUM AWG 14) OR TAE SHALL BE BURIED WITH THE PLASTIC PIPE TO FACILITATE LOCATING. ONE (1) END SHALL BE BROUGHT ABOVE GROUND AT A BUILDING WALL OR RISER. [NFPA 54:1.133] GPG 1211.13

VENTS:

- VENT TERMINATION SHALL BE NOT LESS THAN 10 ' FEET FROM OR AT LEAST 3 ' FEET ABOVE AIR INLET PER CPC 2019.
- PROVIDE DRYER VENT TO OUTSIDE PER UMC 2019.
- GAS VENTS AND NONCOMBUSTIBLE PIPING IN WALLS,PASSING THROUGH THREE FLOORS OR LESS, SHALL BE EFFECTIVELY DRAFT STOPPED AT EACH FLOOR OR CEILING.(CBC 2019),
- PROVIDE A SEPARATE VENT FOR EACH WASTE BRANCH LINE EXCEEDING 15 FEET IN LENGTH.
- THE MINIMUM AREA OF ANY VENT INSTALLED IN A COMBINATION WASTE AND VENT SYSTEM SHALL BE AT LEAST 2' GROSS SECTIONAL AREA OF THE DRAINPIPE SERVED.

E X E C U T I O N

SCOPE OF WORK:

1. PROVIDE ALL LABOR, MATERIAL, EQUIPMENT, FACILITIES. TRANSPORTATION, FEES AND SERVICES NECESSARY FOR A COMPLETE PLUMBING SYSTEM(S) AS INDICATED ON THE DRAWINGS AND SPECIFIED HEREIN. WORKMANSHIP SHALL BE COMPLETE IN EVERY ASPECT, TESTED, APPROVED AND SATISFACTORY TO THE ARCHITECT/ENGINEER AND IN ACCORDANCE WITH LOCAL, STATE AND FEDERAL LAWS HAVING JURISDICTION.

2. IT IS THE DECLARED AND ACKNOWLEDGED INTENT OF THESE SPECIFICATIONS TO PROVIDE A COMPLETE PLUMBING SYSTEM(S), INCLUSIVE OF ALL REQUIRED PARTS AND ACCESSORIES COMPLETE AND READY FOR USE AS DESCRIBED, BUT NOT LIMITED TO THE FOLLOWING:

   2.1. DOMESTIC WATER SERVICE AND DISTRIBUTION.
   2.2. DOMESTIC HOT WATER.
   2.3. SANITARY DRAINAGE SYSTEM.
   2.4. PLUMBING FIXTURES.
   2.5. PLUMBING EQUIPMENT.
   2.6. PLUMBING SPECIALTIES.

3. THE DRAWINGS INDICATE DIAGRAMMATICALLY THE EXTENT AND LOCATION OF THE WORK INCLUDED. WORK INDICATED, BUT HAVING MINOR DETAILS OBVIOUSLY OMITTED, SHALL BE PROVIDED, INCLUDING THESE DETAILS, WITHOUT EXTRA COST TO THE OWNER.

DISINFECTION OF POTABLE WATER SYSTEM:

1. DISINFECT AND FLUSH POTABLE WATER SYSTEM PER LOCAL PLUMBING, BUILDING, AND HEALTH DEPARTMENT REQUIREMENTS.

FLUSHING:

1. CHLORINATED WATER SUPPLY: IF WATER SUPPLY SERVING SYSTEM IS CHLORINATED FROM COMMUNITY WATER SYSTEM, SYSTEM SHALL BE FLUSHED WITH CLEAN, POTABLE WATER UNTIL NO DIRTY WATER APPEARS AT THE POINT OF OUTLET.

2. NON-CHLORINATED WATER SUPPLY: WATER SUPPLY SYSTEM SHALL BE FLUSHED WITH CLEAN, POTABLE WATER UNTIL NO DIRTY WATER APPEARS AT THE POINT OF OUTLET.

LABEL AND IDENTIFICATION:

1. IDENTIFICATION FOR ALL PIPING SYSTEM(S) SHALL COMPLY WITH ANSI A13.1 FOR SIZE OF LETTERING AND BACKGROUND COLOR FIELD.

2. PIPING SYSTEM(S): IDENTIFICATION SHALL INCLUDE THE CONTENTS OF THE PIPING SYSTEM(S) AND AN ARROW INDICATING THE DIRECTION OF FLOW. HAZARDOUS PIPING SYSTEM(S) SHALL ALSO CONTAIN INFORMATION ADDRESSING THE NATURE OF THE HAZARD. IDENTIFICATION SHALL BE REPEATED AT MAXIMUM INTERVALS OF TWENTY-FIVE (25) FEET AND AT EACH POINT WHERE PIPING PASSES THROUGH A WALL, FLOOR OR ROOF. COLOR OF THE PIPE IDENTIFICATION SHALL BE DISCERNIBLE AND CONSISTENT THROUGHOUT THE BUILDING.

3. EQUIPMENT: IDENTIFICATION SHALL INCLUDE SYSTEM NUMBER, CAPACITY, FLOW, RATE, STATIC PRESSURE, PUMP HEAD, HORSEPOWER, VOLTAGE, ETC.

4. VALVE TAGS: PROVIDE BRASS VALVE TAGS AND BRASS "S" HOOK FASTENERS WITH VALVE NUMBER AND TYPE OF SERVICE NOTED ON TAG. PROVIDE DUPLICATE CHARTS, THE CHART SHALL BE FOR ALL VALVES AND SHALL INDICATE VALVE IDENTIFICATION NUMBER, LOCATION AND PURPOSE.

RECORD DRAWING SUBMITTALS:

AT PROJECT CLOSE-OUT, CONTRACTOR SHALL SUBMIT RECORD DRAWINGS (CERTIFIED OR APPROVED) ALSO KNOWN AS "AS-BUILT" DRAWINGS. LAYOUT SUBMITTALS SHALL BE SUBMITTED FOR RECORDS PRIOR TO FINAL ACCEPTANCE IN ELECTRONIC FORM ON COMPACT DISC(S), USING AUTOCAD VERSION 2010 "DWG" FORMAT OR ADOBE ACROBAT "PDF" FORMAT VIEWABLE FROM ADOBE. RECORD DATA EQUIPMENT, FIXTURE AND RELATED APPURTENANCES MAY BE SUBMITTED AS BOUND HARDCOPY OR ELECTRONIC ADOBE ACROBAT "PDF" FORMAT.

2. PROVIDE RECORD SUBMITTALS AS FOLLOWS:
   2.1. OWNER: 1 COPY.
   2.2. ARCHITECT OF RECORD: 1 COPY.
   2.3. ENGINEER OF RECORD: 1 COPY.

3. CONTRACTOR IS NOT ALLOWED TO USE THE CONTRACT DOCUMENTS FOR "AS-BUILT" DRAWINGS. BACKGROUNDS SHALL BE IN THE LATEST RELEASE OF AUTOCAD.

ENGINEER STAMP & SEAL:


DRAWINGS BY:
STUDIO TEN DESIGNS
JASON ALBERT - 678.390.4655
JASON@STUDIOTENDESIGNS

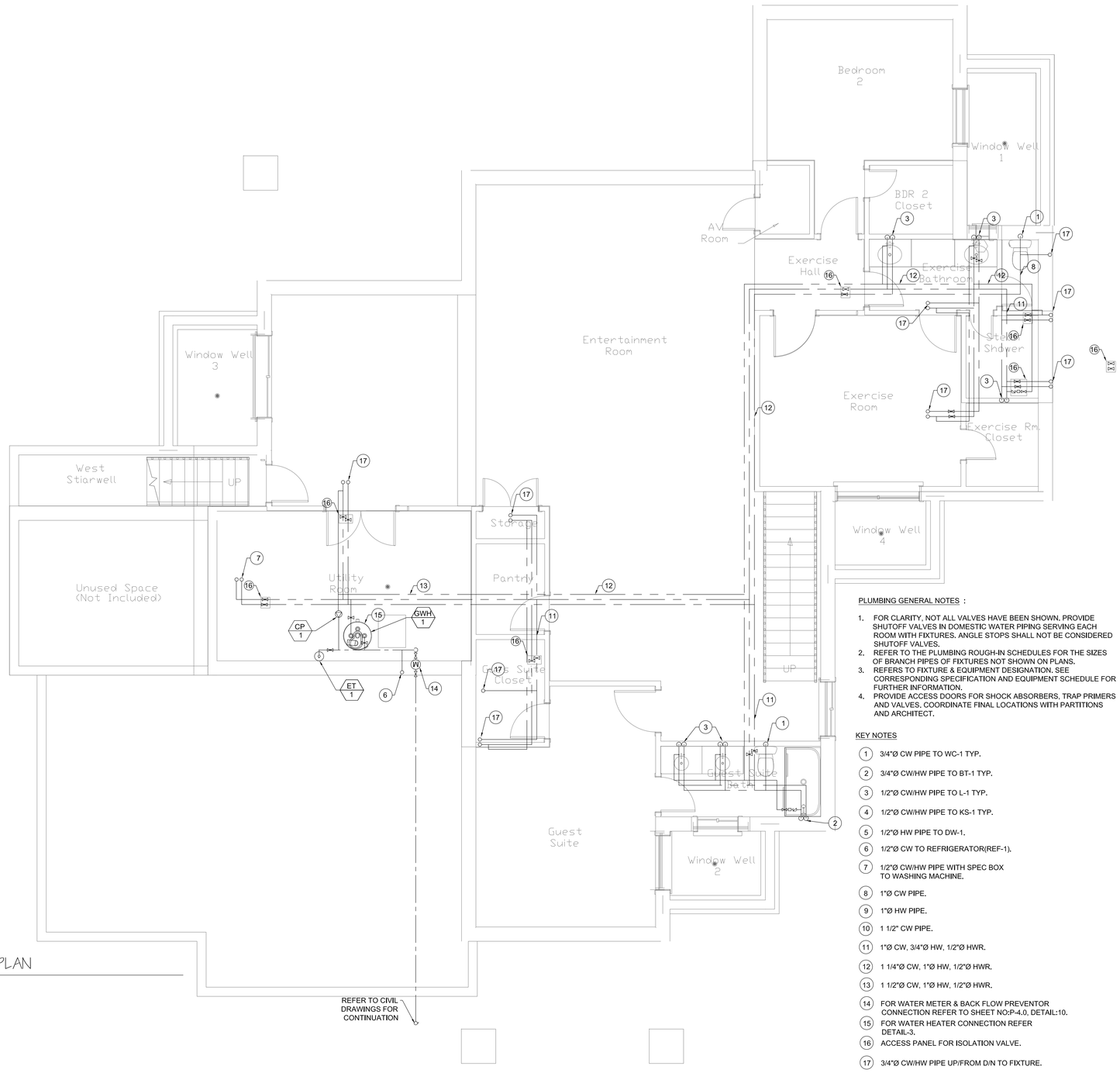

NEW CONSTRUCTION @:
Peer Residence
829 W. Mulberry Street - Louisville, CO 80027

PROJECT MANAGER
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DRAWN BY:
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DOCUMENT PHASE:
RELEASED FOR CONSTRUCTION
NOVEMBER 13, 2022

SHEET TITLE:
GENERAL NOTES AND SYMBOL
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PG.1




## PROPOSED BASEMENT FLOOR PLAN

SCALE: 1/4" = 1'-0"

PLUMBING GENERAL NOTES :

1. FOR CLARITY, NOT ALL VALVES HAVE BEEN SHOWN. PROVIDE SHUTOFF VALVES IN DOMESTIC WATER PIPING SERVING EACH ROOM WITH FIXTURES. ANGLE STOPS SHALL NOT BE CONSIDERED SHUTOFF VALVES.
2. REFER TO THE PLUMBING ROUGH-IN SCHEDULES FOR THE SIZES OF BRANCH PIPES OF FIXTURES NOT SHOWN ON PLANS.
3. REFERS TO FIXTURE & EQUIPMENT DESIGNATION. SEE CORRESPONDING SPECIFICATION AND EQUIPMENT SCHEDULE FOR FURTHER INFORMATION.
4. PROVIDE ACCESS DOORS FOR SHOCK ABSORBERS, TRAP PRIMERS AND VALVES. COORDINATE FINAL LOCATIONS WITH PARTITIONS AND ARCHITECT.

KEY NOTES

1. 3/4"Ø CW PIPE TO WC-1 TYP.
2. 3/4"Ø CW/HW PIPE TO BT-1 TYP.
3. 1/2"Ø CW/HW PIPE TO L-1 TYP.
4. 1/2"Ø CW/HW PIPE TO KS-1 TYP.
5. 1/2"Ø HW PIPE TO DW-1.
6. 1/2"Ø CW TO REFRIGERATOR(REF-1).
7. 1/2"Ø CW/HW PIPE WITH SPEC BOX TO WASHING MACHINE.
8. 1"Ø CW PIPE.
9. 1"Ø HW PIPE.
10. 1 1/2" CW PIPE.
11. 1"Ø CW, 3/4"Ø HW, 1/2"Ø HWR.
12. 1 1/4"Ø CW, 1"Ø HW, 1/2"Ø HWR.
13. 1 1/2"Ø CW, 1"Ø HW, 1/2"Ø HWR.
14. FOR WATER METER & BACK FLOW PREVENTOR CONNECTION REFER TO SHEET NO:P-4.0, DETAIL:10.
15. FOR WATER HEATER CONNECTION REFER DETAIL-3.
16. ACCESS PANEL FOR ISOLATION VALVE.
17. 3/4"Ø CW/HW PIPE UP/FROM D/N TO FIXTURE.

ENGINEER STAMP & SEAL:

DRAWINGS BY:
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NEW CONSTRUCTION @:
Peer Residence
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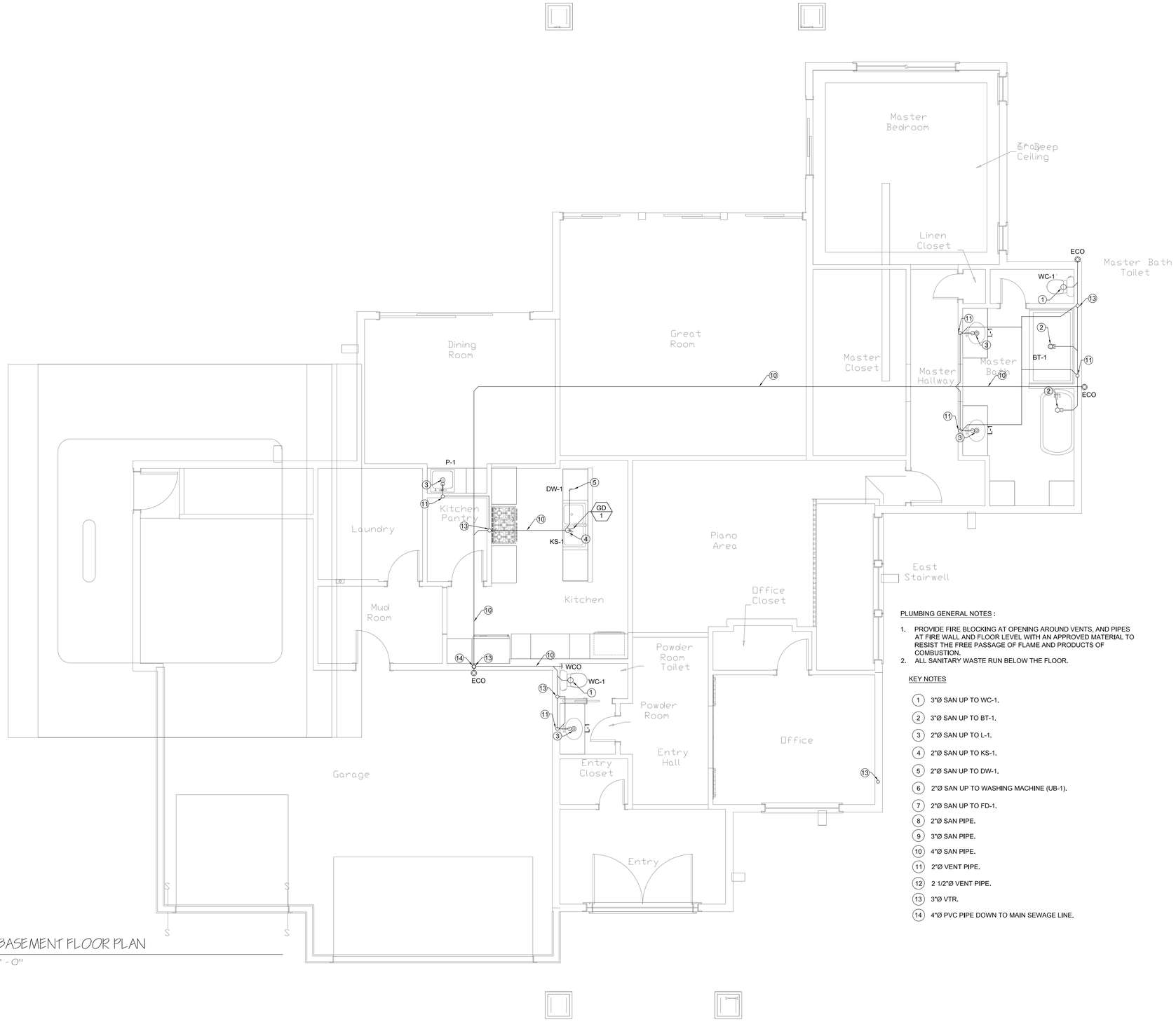
NOVEMBER 13, 2022

SHEET TITLE:

WATER SUPPLY BASEMENT FLOOR PLAN

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P1.0




PROPOSED BASEMENT FLOOR PLAN

SCALE: 1/4" = 1' - 0"

PLUMBING GENERAL NOTES :

1. PROVIDE FIRE BLOCKING AT OPENING AROUND VENTS, AND PIPES AT FIRE WALL AND FLOOR LEVEL WITH AN APPROVED MATERIAL TO RESIST THE FREE PASSAGE OF FLAME AND PRODUCTS OF COMBUSTION.
2. ALL SANITARY WASTE RUN BELOW THE FLOOR.

KEY NOTES

1. 3"Ø SAN UP TO WC-1.
2. 3"Ø SAN UP TO BT-1.
3. 2"Ø SAN UP TO L-1.
4. 2"Ø SAN UP TO KS-1.
5. 2"Ø SAN UP TO DW-1.
6. 2"Ø SAN UP TO WASHING MACHINE (UB-1).
7. 2"Ø SAN UP TO FD-1.
8. 2"Ø SAN PIPE.
9. 3"Ø SAN PIPE.
10. 4"Ø SAN PIPE.
11. 2"Ø VENT PIPE.
12. 2 1/2"Ø VENT PIPE.
13. 3"Ø VTR.
14. 4"Ø PVC PIPE DOWN TO MAIN SEWAGE LINE.

ENGINEER STAMP & SEAL:


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JASON@STUDIOTENDESIGNS


NEW CONSTRUCTION @:
Peer Residence
829 W. Mulberry Street - Louisville, CO 80027

PROJECT MANAGER
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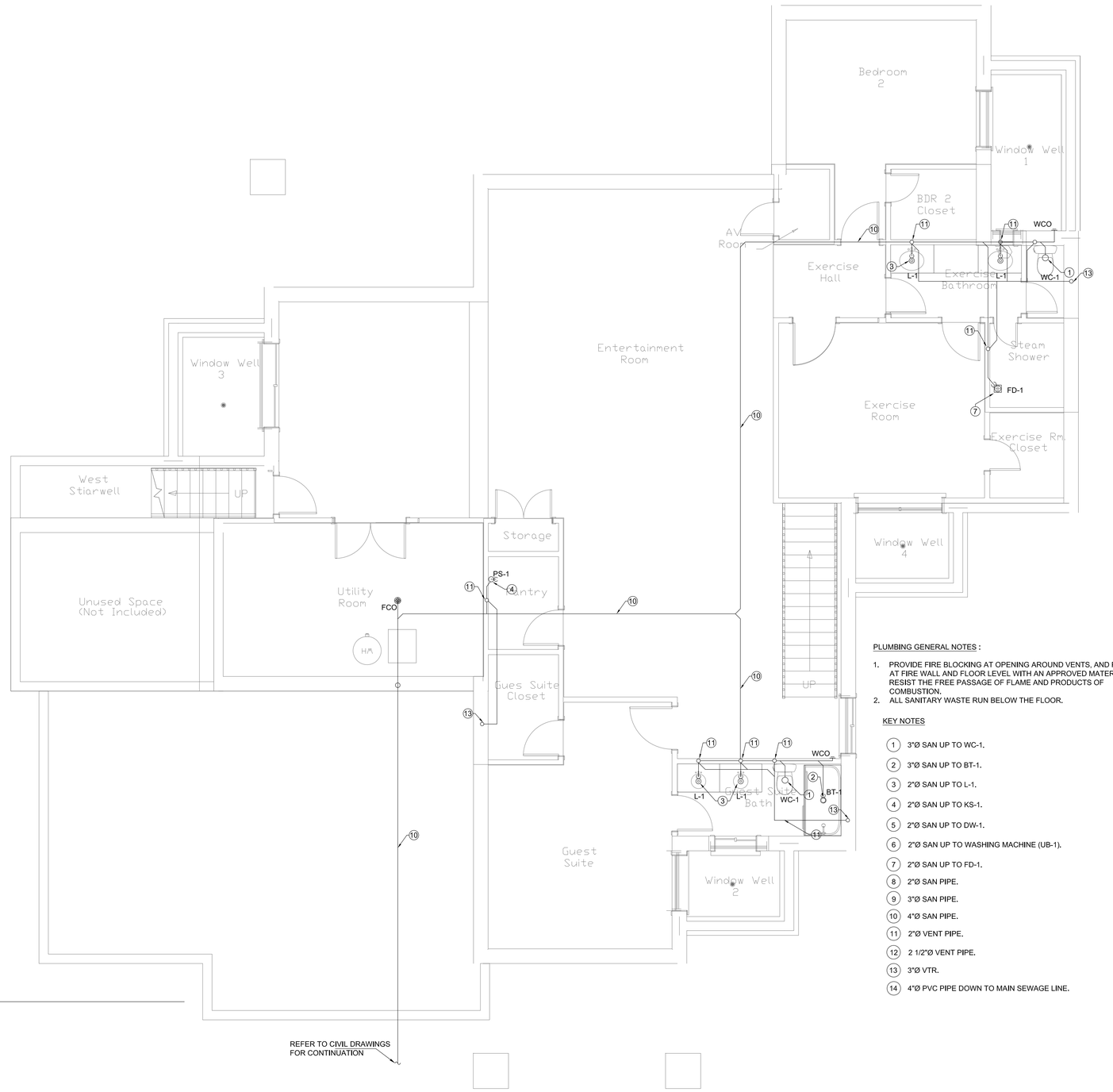
NOVEMBER 13, 2022

SHEET TITLE:

WATER SUPPLY MAIN FLOOR PLAN

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P1.0A




## PROPOSED MAIN FLOOR PLAN

SCALE: 1/4" = 1'-0"

PLUMBING GENERAL NOTES :

1. PROVIDE FIRE BLOCKING AT OPENING AROUND VENTS, AND PIPES AT FIRE WALL AND FLOOR LEVEL WITH AN APPROVED MATERIAL TO RESIST THE FREE PASSAGE OF FLAME AND PRODUCTS OF COMBUSTION.
2. ALL SANITARY WASTE RUN BELOW THE FLOOR.

KEY NOTES

1. 3"Ø SAN UP TO WC-1.
2. 3"Ø SAN UP TO BT-1.
3. 2"Ø SAN UP TO L-1.
4. 2"Ø SAN UP TO KS-1.
5. 2"Ø SAN UP TO DW-1.
6. 2"Ø SAN UP TO WASHING MACHINE (UB-1).
7. 2"Ø SAN UP TO FD-1.
8. 2"Ø SAN PIPE.
9. 3"Ø SAN PIPE.
10. 4"Ø SAN PIPE.
11. 2"Ø VENT PIPE.
12. 2 1/2"Ø VENT PIPE.
13. 3"Ø VTR.
14. 4"Ø PVC PIPE DOWN TO MAIN SEWAGE LINE.

ENGINEER STAMP & SEAL:


DRAWINGS BY:
STUDIO TEN DESIGNS

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JASON@STUDIOTENDESIGNS


NEW CONSTRUCTION @:
Peer Residence
829 W. Mulberry Street - Louisville, CO 80027

PROJECT MANAGER
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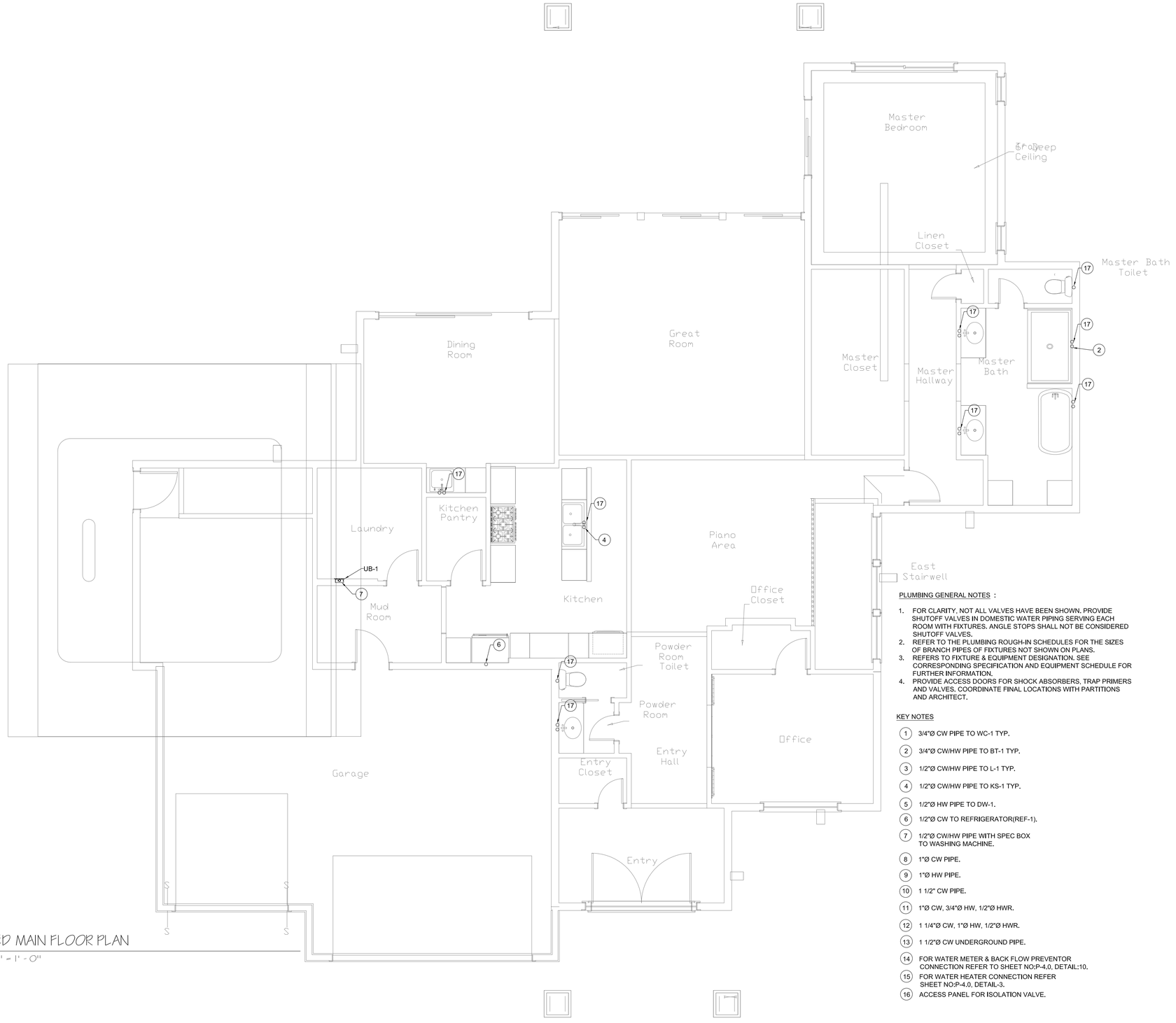
NOVEMBER 23, 2022

SHEET TITLE:

DRAINAGE BASEMENT FLOOR PLAN

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P2.0




1 PROPOSED MAIN FLOOR PLAN

SCALE: 1/4" = 1'-0"

PLUMBING GENERAL NOTES :

1. FOR CLARITY, NOT ALL VALVES HAVE BEEN SHOWN. PROVIDE SHUTOFF VALVES IN DOMESTIC WATER PIPING SERVING EACH ROOM WITH FIXTURES. ANGLE STOPS SHALL NOT BE CONSIDERED SHUTOFF VALVES.
2. REFER TO THE PLUMBING ROUGH-IN SCHEDULES FOR THE SIZES OF BRANCH PIPES OF FIXTURES NOT SHOWN ON PLANS.
3. REFERS TO FIXTURE & EQUIPMENT DESIGNATION. SEE CORRESPONDING SPECIFICATION AND EQUIPMENT SCHEDULE FOR FURTHER INFORMATION.
4. PROVIDE ACCESS DOORS FOR SHOCK ABSORBERS, TRAP PRIMERS AND VALVES. COORDINATE FINAL LOCATIONS WITH PARTITIONS AND ARCHITECT.

KEY NOTES

1. 3/4"Ø CW PIPE TO WC-1 TYP.
2. 3/4"Ø CW/HW PIPE TO BT-1 TYP.
3. 1/2"Ø CW/HW PIPE TO L-1 TYP.
4. 1/2"Ø CW/HW PIPE TO KS-1 TYP.
5. 1/2"Ø HW PIPE TO DW-1.
6. 1/2"Ø CW TO REFRIGERATOR(REF-1).
7. 1/2"Ø CW/HW PIPE WITH SPEC BOX TO WASHING MACHINE.
8. 1"Ø CW PIPE.
9. 1"Ø HW PIPE.
10. 1 1/2" CW PIPE.
11. 1"Ø CW, 3/4"Ø HW, 1/2"Ø HWR.
12. 1 1/4"Ø CW, 1"Ø HW, 1/2"Ø HWR.
13. 1 1/2"Ø CW UNDERGROUND PIPE.
14. FOR WATER METER & BACK FLOW PREVENTOR CONNECTION REFER TO SHEET NO:P-4.0, DETAIL:10.
15. FOR WATER HEATER CONNECTION REFER SHEET NO:P-4.0, DETAIL-3.
16. ACCESS PANEL FOR ISOLATION VALVE.

ENGINEER STAMP & SEAL:


DRAWINGS BY:
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NEW CONSTRUCTION @:
Peer Residence
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PROJECT MANAGER
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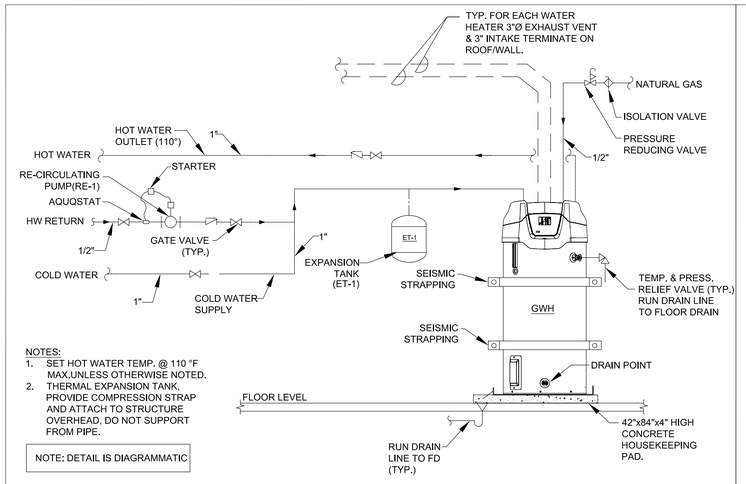
NOVEMBER 13, 2022

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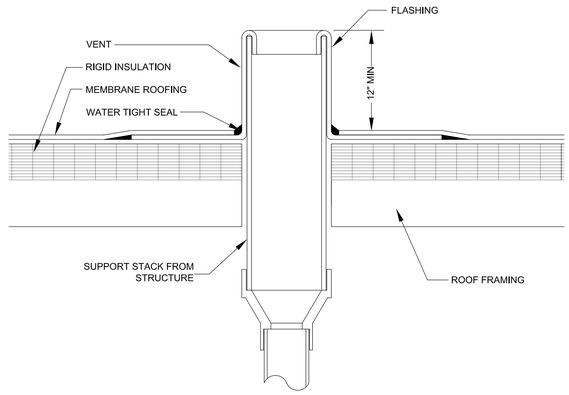
DRAINAGE MAIN FLOOR PLAN

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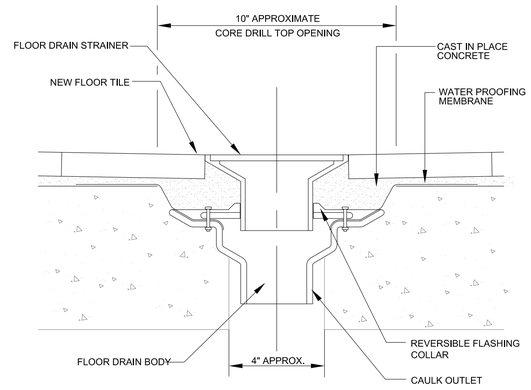
P2.0A




GAS WATER HEATER CONNECTION DETAIL — 3
NOT TO SCALE

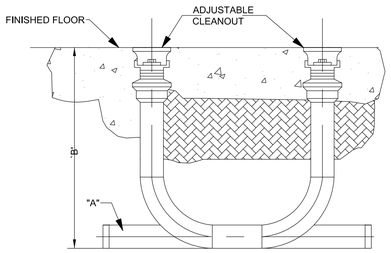



VENT THROUGH ROOF DETAIL — 2
NOT TO SCALE

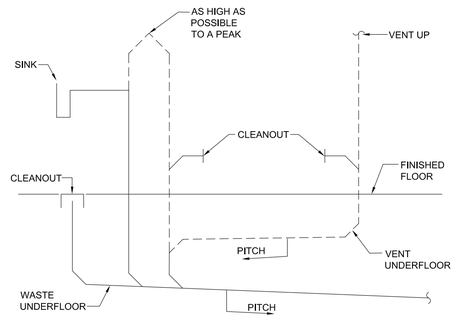



FLOOR DRAIN DETAIL — 1
NOT TO SCALE

| MIN.DIMENSIONS "A" = PIPE SIZE `B' = PIPE INVERT | | |
|---|---|---|
| "A" | 4" | 6" |
| `B' | 1'-8 1/2" | 1'-10 3/4" |

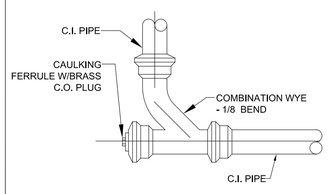



TWO WAY CLEANOUT TO FINISHED FLOOR — 8
NOT TO SCALE

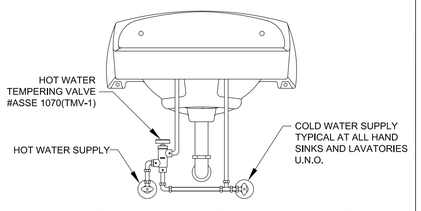



NOTES:
1. PROVIDE DRAINAGE FITTINGS ON ALL SECTIONS OF THE VENT UNDER FLOOR AND PROVIDE MINIMUM SLOPE OF 1/4" PER FOOT TOWARD DRAINAGE POINT.

ISLAND VENTING DETAILS — 7
NOT TO SCALE

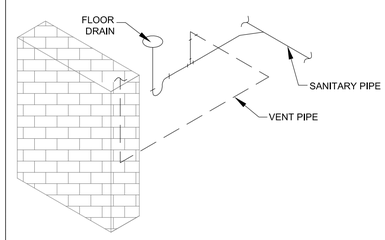



EXPOSED CLEANOUT DETAILS — 6
NOT TO SCALE

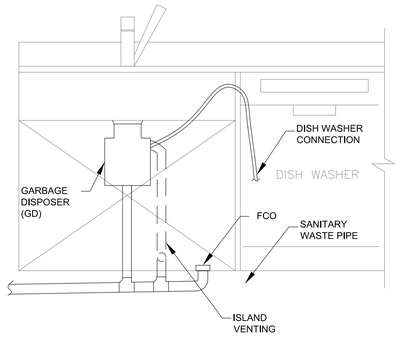



NOTE: PROVIDE TRAP AND SUPPLY INSULATION AS REQUIRED. SEE SPECIFICATIONS

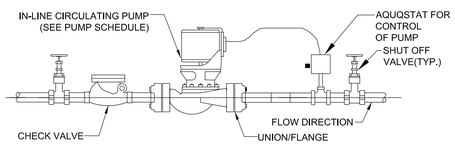
LAVATORY AND HAND SINK TEMPERING VALVE DETAIL — 5
NOT TO SCALE



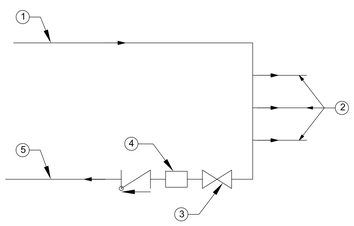

VENTING DETAIL — 4
NOT TO SCALE




GARBAGE DISPOSER AND DISHWASHER DRAINAGE CONNECTION DETAILS — 13
NOT TO SCALE

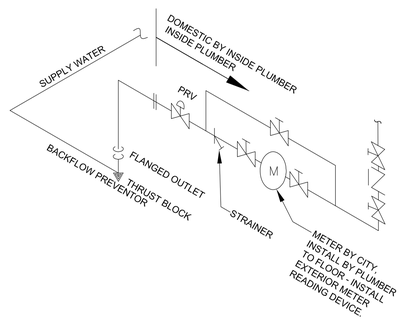



HOT WATER RECIRCULATION PUMP DETAILS — 12
NOT TO SCALE



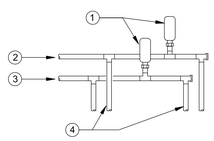

(#) KEY NOTES:
1. HOT WATER SUPPLY LINE FROM HOT WATER HEATER/GENERATOR.
2. PLUMBING FIXTURES.
3. ISOLATION VALVE.
4. FLOW CONTROL VALVE (REFER PLAN FOR SET FLOW)
5. HOT WATER RETURN LINE TO HOT WATER HEATER/GENERATOR.

HOT WATER RETURN & FLOW CONTROL DEVICE CONNECTION DETAIL — 11
NOT TO SCALE




DOMESTIC WATER METERING STATION DETAIL — 10
NOT TO SCALE

(#) KEY NOTES:
1. WATER HAMMER ARRESTOR (WHA) INSTALL AHEAD OF LAST FIXTURE SERVED BY BRANCH. (TYP)
2. CW PIPE
3. HW PIPE
4. CONNECT TO FIXTURE




| WATER ARRESTOR SCHEDULE | | |
|---|---|---|
| P.I.D. SYMBOL | FIXTURE UNIT RATING | SIZE (MM) |
| A | 1-11 | 25 |
| B | 12-32 | 32 |
| C | 33-60 | 40 |
| D | 61-113 | 50 |

WATER HAMMER ARRESTER DETAIL — 9
NOT TO SCALE

ENGINEER STAMP & SEAL:

DRAWINGS BY:
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DOCUMENT PHASE:
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NOVEMBER 13, 2022

SHEET TITLE:
PLUMBING DETAILS

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P3.0

## PLUMBING FIXTURE SCHEDULE

| SYMBOL | MANUFACTURER | MODEL | FIXTURE | MOUNTING | HW | CW | WASTE | VENT | ACCESSORIES / REMARKS | |
|---|---|---|---|---|---|---|---|---|---|---|
| BFP-1 | WATTS | LF007 | - | - | - | - | - | - | 1 1/2" RPZ, TEMPERATURE RANGE:33°F-180°F, MAXIMUM WORKING PRESSURE:175 PSI, LEAD FREE. |  |
| ECO | ZURN | Z-1400-S | EXTERNAL CLEAN OUT | FLOOR | - | - | SEE REMARKS | - | MATCH CONNECTED PIPE SIZE UP TO 4". CAST IRON BODY WITH BOTTOM OUTLET. |  |
| L-1 | KOHLER | K-2882 | LAVATORY | WALL | 1/2" | 1/2" | 2" | 2" | VITREOUS CHINA, UNDER MOUNT, ASME A112.19.2/CSA B45.1, ICC/ANSI A117.1, 17-1/4"(LENGTH)X 13"(WIDTH) X 3-1/8"(WATER DEPTH). |  |
| BT-1 | KOHLER | K-6368-0 | BATH TAB | FLOOR | 3/4" | 3/4" | 3" | 2" | FREESTANDING DESIGN CREATE A STRIKING FOCAL POINT, ACRYLIC MATERIAL, CENTRAL DRAIN, WATER CAPACITY-72.6 GAL., WEIGHT-110 LBS. |  |
| WC-1 | KOHLER | K3999 | WATER CLOSET | FLOOR | - | 3/4" | 3" | 2" | WATER CLOSET: ELONGATED BOWL, TWO-PIECE DESIGN, SINGLE-FLUSH GRAVITY FORCE. 1.28 GALLON FLUSH. |  |
| WHA | PRECISION PLUMBING PRODUCTS | SC SERIES | WATER HAMMER ARRESTOR | - | - | - | - | - | INSTALL AND SIZE PER MANUFACTURER'S INSTRUCTIONS. INSTALL AT ENDS OF COLD AND HOT WATER MAIN BRANCHES AND AT FAST-CLOSING VALVES OR ONE AT EACH FIXTURE. SHOW LOCATIONS ON RECORD DRAWINGS. |  |
| BALL VALVES | APOLLO / COBRANCO OR EQUAL | - | BALL VALVES | - | - | - | - | - | FULL PORT, BRASS BODY, 600 WOG |  |
| PIPE HANGERS | B-LINE OR EQUAL | 200F | PIPE HANGERS | - | - | - | - | - | - |  |
| KS-1 | KOHLER | K-5871-4A2 | KITCHEN SINK | WALL | 1/2" | 1/2" | 2" | 2" | TOP-MOUNT WORKSTATION KITCHEN SINK, CAST IRON MATERIAL, 33" WIDTH AND 9" DEPTH SINGLE BOWL. |  |
| WCO | ZURN | Z-1447 | WALL CLEAN OUT | WALL | - | - | SEE REMARKS | - | MATCH CONNECTED PIPE SIZE UP TO 4". SECONDARY CLOSURE PLUG |  |
| UB-1 | SHARKBITE | SKU25099 | WASHING MACHINE UTILITY BOX | - | 1/2" | 1/2" | 2" | - | UB BOX CAN BE USED ON COPPER,CPVC, PEX AND PE-RT. MAXI. WORKING PRESSURE: 200 PSI., MAXI. TEMPERATURE: 200°F, LEAD FREE DZR BRASS. |  |
| W | SENSUS | WDS-10017-07 | WATER METER | - | - | 1 1/2" | - | - | NOMINAL DIAMETER (DN): 1 1/2", MAXIMUM OPERATING PRESSURE: 200 PSI, FLOW RATE: 0.5 - 200 GPM |  |

NOTE:
1. REFER TO ARCHITECTURE, OWNERSHIP OR INTERIOR DESIGNER FOR ALL FIXTURE SELECTION AND ACCESSORIES. ALL ARE SUBJECT TO CHANGE.

### GAS WATER HEATER SCHEDULE

| SYMBOL | QTY. | LOCATION | TANK CAPACITY (GAL) | RATED CAPACITY (GAL) | GAS INPUT THOUS. BTU/HR | TEMPERATURE RISE (°F) | RECOVERY IN G.P.H. 90°F RISE | WATER CONNECTION SIZE (INCH.) | SUPPLY GAS CONNECTION (INCH.) | VENT SIZE (INCH.) | ELECTRIC DATA | | DIMENSION ( INCH.) | | SHIPPING WEIGHT (LBS) | UNIFORM ENERGY FACTOR (UEF) | MODEL | BASIS OF DESIGN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | FLA | V/PH/HZ | DIAMETER | HEIGHT | | | | |
| GWH-1 | 1 | UTLITY ROOM | 40 | 38 | 36 | 90 | 36.3 | 3/4" | 1/2" | 4"Ø | 5.0 | 120/1/60 | 19-1/2" | 63-1/4" | 135 | 0.66 | XP40T06HE36U0 | RHEEM |

NOTES:
1. INSTALL PER MANUFACTURER INSTRUCTIONS.
2. USE ONLY VENT TERMINALS PROVIDED OR FACTORY AUTHORIZED TERMINALS FOR VENTING THIS WATER HEATER.
3. THIS WATER HEATER REQUIRES IT'S OWN SEPARATE VENTING SYSTEM. DO NOT CONNECT THE EXHAUST VENT INTO ANY EXITING VENT OR CHIMNEY.

### RECIRCULATING PUMP SCHEDULE

| SYMBOL | DESCRIPTION | LOCATION | QTY. | FLOW (GPM) | HEAD (FEET) | FLANGE SIZE NPT(IN) | ELECTRICAL | | | | | WEIGHT (LBS) | MODEL | BASIS OF DESIGN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | HP | VOLTAGE | PHASE | RPM | F.L.A (AMPS) | | | |
| RE-1 | RECIRCULATING PUMP | UTLITY ROOM | 1 | 5 | 8 | 3/4 | 1/12 | 120 | 1 | 1725 | 1.75 | 21 | SERIES 100 | BELL & GOSSETT |

NOTES:
1. PROVIDE L-6007-A HONEYWELL, INC.OR APPROVED EQUAL, AQUAST INSTALLED IN HOT WATER CIRCULATING LINE ADJACENT TO PUMP TO STOP PUMP AT SET TEMPERATURE IN FIELD.

### EXPANSION TANK SCHEDULE

| SYMBOL | QTY. | TANK VOLUME (GAL) | MAX. ACCEPTANCE FACTOR | TANK ACCEPT. VOLUME (GAL) | TYPE | BLADDER MATERIAL | SHELL MATERIAL | MAX. OPERATING TEMPERATURE (°F) | MAX. WORKING PRESSURE (PSIG) | SHIPPING WEIGHT (LBS) | DIMENSION (DIA. X HIEGHT) | MODEL | BASIS OF DESIGN | NOTES |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ET-1 | 1 | 2.0 | 0.45 | 0.9 | REPLACEBLE BLADDER | HEAVY DUCTY BUTYL | STEEL | 200 | 150 | 5 | 8"X13" | ST-5 | AMTROL | 1,2,3 |

NOTES:
1. INSTALL PER MANUFACTURER INSTRUCTIONS.
2. PROVIDE WITH STANDARD SCHRADER TIRE VALVE CONNECTION.
3. ADJUST CHARGE PRESSURE IN FIELD.

### PIPING MATERIALS SCHEDULE

| PLAN TAG | PLUMBING SYSTEM | | | SYSTEM MATERIAL AND FITTING SPECIFICATION |
|---|---|---|---|---|
| | DESCRIPTION | INSTALLATION | SIZES | |
| SW | SANITARY (SOIL) PIPING | SUSPENDED | 2-1/2" AND SMALLER | SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL |
| | | | 3" AND LARGER | SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL |
| | | BURIED | 3" AND LARGER | SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL |
| V | VENT PIPING (ALL SYSTEMS) | SUSPENDED | 2-1/2" AND SMALLER | SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL |
| | | | | SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL |
| | | | 3" AND LARGER | SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL |
| | | | | SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL |
| | | BURIED | 2" AND LARGER | SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL |
| | | | | SANITARY PIPE AND FITTINGS SHALL BE POLYVINYL CHLORIDE (PVC) PER ASTM D 2665, WITH GASKETS PER ASTM C 1440, ELASTOMERIC SEAL |
| CW | DOMESTIC COLD WATER DISTRIBUTION | SUSPENDED | 2-1/2" AND SMALLER | COPPER PIPE, ASTM B88, DRAWN TYPE L AND K WITH WROUGHT COPPER PRESSURE FITTINGS, SOLDERED, ASTM B16.22 |
| | | | 3" AND LARGER | COPPER PIPE, ASTM B88, DRAWN TYPE L AND K WITH WROUGHT COPPER PRESSURE FITTINGS, SOLDERED, ASTM B16.22 |
| | | BURIED | 3" AND LARGER | COPPER PIPE, ASTM B88, DRAWN TYPE L AND K WITH WROUGHT COPPER PRESSURE FITTINGS, SOLDERED, ASTM B16.22 |
| HW | DOMESTIC HOT WATER DISTRIBUTION | SUSPENDED | 2-1/2" AND SMALLER | COPPER PIPE, ASTM B88, DRAWN TYPE L AND K WITH WROUGHT COPPER PRESSURE FITTINGS, SOLDERED, ASTM B16.22 |
| | | | 3" AND LARGER | COPPER PIPE, ASTM B88, DRAWN TYPE L AND K WITH WROUGHT COPPER PRESSURE FITTINGS, SOLDERED, ASTM B16.22 |
| | | BURIED | - | COPPER PIPE, ASTM B88, DRAWN TYPE L AND K WITH WROUGHT COPPER PRESSURE FITTINGS, SOLDERED, ASTM B16.22 |

NOTE:
"HE MATERIALS ARE SUBJECT TO CHANGE WITH APPROVAL OF GENERAL CONTRACOTR OR SOMETHING SIMILAR.

### GARBAGE DISPOSER SCHEDULE

| SYMBOL | LOCATION | FEED TYPE | QTY. | HORSE POWER (HP) | RPM | DRAIN CONNECTION (INCH.) | ELECTRIC DATA | | | | WEIGHT (LBS) | MODEL | MANUFACTURER |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | AMPS | VOLTAGE | PHASE | HZ | | | |
| GD-1 | KITCHEN | CONTINUOUS | 1 | 1/2 | 2600 | 2" | 4.5 | 115V | 1 | 60 | 7.75 | GXP50C | MOEN |

### THERMOSTATIC MIXING VALVE

| TAG NO. | DESCRIPTION | CONNECTION | | MODEL | MANUFACTURER | COMMENTS |
|---|---|---|---|---|---|---|
| | | COLD | HOT | | | |
| TMV-1 | THERMOSTATIC MIXING VALVE (LOCAL) | 3/8" | 3/8" | LFUSG-B | WATTS | ASSE #1070 |
| TMV-2 | THERMOSTATIC MIXING VALVE (BATHTUB) | 1/2" | 1/2" | BILTMORE SERIES 900 | WATTS | ASSE #1016 |

NOTES:
1. INSTALL PER MANUFACTURER INSTRUCTIONS.

### HOT WATER DEMAND CALCULATIONS UNIT- A

| MARK | QTY | FIXTURE/EQUIPMENT | INDIVIDUAL DEMAND GAL/HR/FIXTURE (140 °F) | TOTAL DEMAND GAL/HR/FIXTURE (140 °F.) |
|---|---|---|---|---|
| | | GENERAL SERVICE LOAD: | | |
| L-1 | 7 | LAVATORY - PRIVATE | 2 | 14 |
| KS-1 | 2 | KITCHEN SINK | 10 | 20 |
| DW-1 | 1 | DISH WASHER | 15 | 15 |
| SH-1 | 1 | SHOWER | 30 | 30 |
| BT-1 | 2 | BATH TUB | 20 | 40 |
| SK-1 | 1 | SINK | 15 | 15 |
| WM-1 | 1 | WASHING MACHINE (LAUNDRY) | 20 | 20 |
| | | | | |
| | | TOTAL GENERAL SERVICE DEMAND (140 °F) | | 154 |
| | | | | |
| | | GENERAL SERVICE LOAD (PRIVATE RESIDENCE CLASSIFICATION): | | |
| | | TOTAL DEMAND (GPH) (140 °F) | 0.30 FACTOR: | 46.2 |
| | | TOTAL STORAGE (GALLON) (140 °F) | 0.70FACTOR: | 32.34 |
| | | | | |
| | | TOTAL NATURAL GAS DEMAND(BTU/HR) | | 34636 |
| | | TOTAL NATURAL GAS DEMAND(CFT/HR) | | 36 |

UNIT : PLUMBING FIXTURE DEMAND TABULATION

| FIXTURE | DESCRIPTION | OCCUPANCY | QTY. | DRAINAGE FIXTURE UNITS | SUB-TOTAL | LOAD VALUES IN WATER (EACH) SUPPLY FIXTURE UNITS (WSFU) | | | LOAD VALUES IN WATER (TOTAL) SUPPLY FIXTURE UNITS (WSFU) | | | REMARK |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | COLD | HOT | TOTAL | COLD | HOT | TOTAL | |
| DW-1 | DISHWASHING MACHINE (AUTOMATIC) | PRIVATE | 1 | 2 | 2 | - | 1.1 | 1.5 | - | 1.1 | 1.5 | |
| KS-1 | KITCHEN SINK | PRIVATE | 2 | 2 | 4 | 1.0 | 1.0 | 1.4 | 2.0 | 2.0 | 2.8 | |
| SH-1 | SHOWER HEAD | PRIVATE | 1 | 2 | 2 | 1.5 | 1.5 | 2.0 | 1.5 | 1.5 | 2 | |
| WM | WASHING MACHINE | PRIVATE | 1 | 3 | 3 | 3.0 | 3.0 | 4.0 | 3.0 | 3.0 | 4 | |
| L-1 | LAVATORY | PRIVATE | 7 | 1 | 7 | 0.8 | 0.8 | 1.0 | 5.3 | 5.3 | 7 | |
| WC-1 | WATER CLOSET, FLUSH TANK (1.6 GPF WATER CLOSET) | PRIVATE | 4 | 3 | 12 | 1.9 | - | 2.5 | 7.5 | - | 10 | |
| BT-1 | BATHTUB | PRIVATE | 2 | 2 | 4 | 3.0 | 3.0 | 4.0 | 6.0 | 6.0 | 8 | |
| FD-1 | FLOOR DRAIN | PRIVATE | 1 | 2 | 2 | 0 | 0 | 0 | 0 | 0 | 0 | |
| CW-1 | CLOTH WASHER | PRIVATE | 1 | 3 | 3 | 1.125 | 1.125 | 1.5 | 1.125 | 1.125 | 1.5 | |
| TOTALS | | | | | 39 | DFU | | | 26.4 | 20.0 | 36.8 | WSFU |
| DFU = DRAINAGE FIXTURE UNITS | | | | | EIGHTH | INCH SLOPE PER FOOT | | | 17 | 14.3 | 22.2 | GPM |
| WSFU = WATER SUPPLY FIXTURE UNITS | | | | | 4" | DIAMETER OF PIPE (INCHES) | | | 1 1/4" | 1 1/4" | 1 1/2" | INCHES REQ'D. |

ENGINEER STAMP & SEAL:


DRAWINGS BY:
STUDIO TEN DESIGNS

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NOVEMBER 13, 2022

SHEET TITLE:
PLUMBING SCHEDULES

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